



Developing APIs with Google Cloud's Apigee API Platform

Welcome to Developing APIs with Apigee. This 3-day instructor-led course introduces you to API Design, and teaches you how to build APIs on Google Cloud's Apigee API Platform.

I ❤️ APIs

cloud.google.com

apigee.com



Introductions

Your instructor + You

Background

Position

Organization

Course goals



Introductions:

- Your instructor + You
- Background
- Position
- Organization

Facilities



Parking



Facilities



Food



Facilities:

- Parking
- Facilities
- Food

Course etiquette



Please silence
your phone and
take calls outside.



Recording
this class
is prohibited.



Ask questions
interactively or
via chat (online).

Course etiquette:

- Recording this class is prohibited.
- Please silence your phone and take calls outside.
- Ask questions interactively or via chat (online).

Target audience and prerequisites

Developers, architects, or engineers responsible for the solutioning, design, or implementation of APIs or digital products that leverage APIs.

- Familiarity with HTTP, XML, JSON, and JavaScript is helpful



This course is targeted at developers, architects, or engineers who want to design, build, or implement APIs on Apigee.

Ideally you have development experience. Familiarity with HTTP, XML, JSON, and JavaScript is helpful but not required.

Course Objectives

Identify the purpose and value of Google Cloud's Apigee API Platform.

Learn how to design REST APIs.

Learn how to build APIs using Apigee.

Identify and use Apigee features that help you secure, scale, and manage APIs and API products.



Let's take a look at the overall objectives of the course. First, you'll learn about Apigee and how it can be used to solve significant problems for enterprises.

You will learn how to design REST APIs, and how to build them using Apigee.

And you will also learn how to leverage Apigee features that allow you secure, scale, and manage your APIs and API products.

To solidify your learning, the course design enables you to learn concepts and then apply them in immersive hands-on labs.

Lab Design

Labs will teach you how to build Apigee API proxies.

Two types of labs:

- Core labs (1, 2, 3,...)
 - Labs build upon each other
- Supplemental labs (2a, 5a, ...)
 - Labs are standalone



The labs in this course teach you how to build Apigee API proxies. API proxies are used to implement APIs on Apigee.

You will be using Google Skills for the labs in this course. Each lab will create a new Apigee X organization for you including the prerequisite assets.

There are two types of labs in this course.

There are 13 core labs. These labs will build upon each other, using the concepts you learn in the class. These labs are numbered 1 through 13. The starting point

There are also 4 supplemental labs. These are standalone labs that explore other concepts you will learn during the class. These labs have a letter as part of the lab number. For example, lab 2a is a supplemental lab introduced at the same time as core lab 2.

When we break for labs, please finish the core lab first. The following labs will build upon previous core labs.

Course Agenda

Day 1	Day 2	Day 3
API Design and Fundamentals	API Security	API Development



The Developing APIs course series is split into three days.

Day 1 focuses on: API Design and Fundamentals

Day 2 focuses on: API Security

Day 3 focuses on: API Development

Agenda - Day 1

API Design and Fundamentals

- 1 Apigee Overview
- 2 API First and OpenAPI Specifications
REST API Design, Part I (Basics)
- 3 API Proxies
Lab 1: Generating an API proxy using an OpenAPI spec
Lab 2: Target servers (+ supplemental lab)
- 4 API Products
REST API Design, Part II (Responses)
Lab 3: Publishing APIs as Products



Let's step through the modules and core labs for the three days.

In day 1, API Design and Fundamentals, you'll begin with an overview of the Apigee platform.

You'll learn about the concept of API First Design, and you will learn about OpenAPI specifications. You will also learn about the basics of REST API design.

Next, you'll learn about API proxies, and complete two core labs and one supplemental lab.

Finally you will learn about API products, you will learn about responses for REST APIs, and you will complete one core lab and one supplemental lab.

That will conclude day 1.

Agenda - Day 2

API Security

- 5 Authentication, Authorization, and OAuth
Lab: Using OAuth client credentials grant type
- 6 Content, Transport, and Platform Security
Lab: JSON threat protection (+ supplemental lab)
Lab: Platform threat protection (+ supplemental lab)
- 7 Mediation
Lab: Add XML support
Lab: Mashup (+ supplemental lab)
Lab: Shared flows
Lab: Fault handling



In day 2, API Security, we'll learn all about API Security concerns, and take a deep dive into OAuth. You will complete a lab that adds OAuth to your proxy. You'll also learn about JWT ("jot") tokens and federated identity for APIs.

You'll also learn about other types of security concerns, including content-based attacks. You will learn how API calls are protected across the network, and you will learn how to protect against internal access of sensitive data. You will complete 2 more core security labs, and 2 supplemental labs.

We will also learn about mediation, including how we call multiple services during an API call. We will learn how to use custom code in your proxies, how you share logic between your proxies, and how errors are handled in Apigee. You will explore these topics using 4 more core labs and one more supplemental lab.

We will probably not get through all of Mediation on day 2, so on day 3 we'll pick up where we left off.

That will end a packed day 2.

Agenda - Day 3

API Development

- 7 Mediation (finish)
- 8 Traffic management
Lab: Traffic management
Lab: Caching
- 9 API Publishing
REST API Design, Part III (Versioning)
Lab: Developer portal
- 10 Logging and Analytics
- 11 Advanced Topics



Finally, day 3 is API Development.

After finishing mediation, we will learn about traffic management, and how to use the spike arrest, quota, and caching policies to manage traffic. You will complete 2 core labs to explore these topics.

Next, you will learn about API publishing, and how you can use an Apigee developer portal to engage app developers, and you will complete a lab to create a developer portal and publish your API on it. You will also learn about versioning your REST APIs.

You will learn how to log errors from your proxies, and how you can use analytics to determine the current state of your APIs and API program.

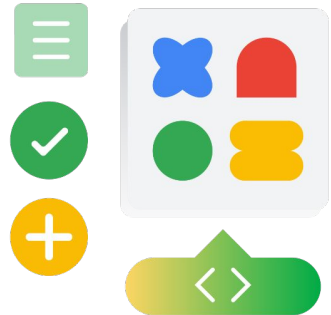
Finally, we will explore the tools available for you to do CI/CD, and you'll learn about the different Apigee deployment models you can leverage.

At the end of day 3, you will have learned and applied the skills to design and develop APIs using Apigee.

Lab Environment

For each lab, Google Skills offers:

- ✓ A free set of resources for a fixed amount of time
- ✓ A clean environment with permissions



Google Skills provisions you with Google account credentials, so you can access the Google Cloud Console for each lab at no cost.

Specifically, for each lab, Google Skills offers:

- A free set of resources for a fixed amount of time
- A clean environment with permissions

Open Google Skills

- 1 Open an incognito window (or private and/or anonymous window).
- 2 Go to the URL your instructor provides.
- 3 Sign In with existing account or Join with new account (with email you used to register for the course).
- 4 Launch the course from the **Dashboard** page.

Access issues

The process to open Google Skills can differ based on credentials used. Please reach out to your instructor if you have any access issues.



Up next

Activities

Classrooms

SCHEDULED COURSE

Developing APIs with Google Cloud's Apigee API Platform

▶ Join

Go ahead and open Google Skills.

1. **Open an incognito window** (or private/anonymous window).
2. **Go** to the Google Skills URL your instructor provides.
3. **Sign in** with an existing account or **join** with a new account (with the email you used to register for the course).
4. Launch the course from **Profile**.

View your labs

LabsLecture Notes

✓ Console and Cloud Shell v1.5

○ Infrastructure Preview v1.5

⚠ Virtual Networking v1.5Lab Currently Disabled

⚠ Bastion Host v1.5Lab Currently Disabled

← Lab Completed

← Active Lab

← Not yet available

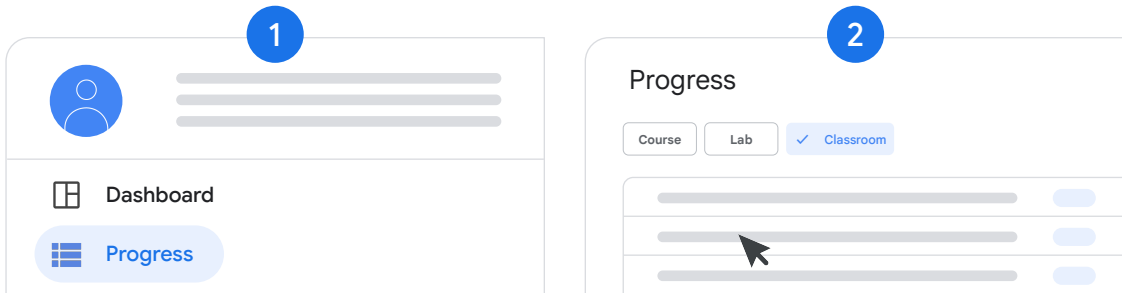
Do NOT launch a lab until instructed to do so!

View lecture notes

Labs	Lecture Notes
00 Course Intro	
01 Introduction to GCP	
02 Virtual Networks	
03 Virtual Machines	
04 Cloud IAM	

End of class: Materials

Materials are available for 2 years



Click on your Profile and select **Progress**.

Select the class from the **Classroom** list.



You can view the course materials within Google Skills as follows:

1. Click on **My account** (top-right of the screen).
2. Select **Progress**.
3. Select the **Classroom** activity filter.
4. Select the class from the **Classroom** list.

Materials are available for 2 years following the completion of a course.

